

Sample Exam

Medical Image Analysis

Student name:_____

Immatriculation number:_____

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions, 2D image slices are acquired in one pre-defined direction one after the other, and the final 3D volume is created by stacking the 2D slices. For instance, in axial acquisitions 2D slices are acquired in the transverse plane, in sagittal acquisitions in the sagittal plane, and in coronal acquisitions in the coronal plane. The voxel sizes of 2D acquisitions are given based on the acquisition direction, as in-plane and through-plane voxel sizes, where the first is represented with two values (pixel size in the imaging plane) and the latter corresponds to a single value (length of the voxel in the third dimension). Recall that the field-of-view is the real-world size of the entire image, i.e. image size $\times$ voxel size. Based on this please compute the following.
 - (a) A *sagittal* acquisition with voxel size in-plane $0.5 \times 0.5 \text{ mm}^2$ with 384 pixels in both directions and voxel size through-plane 4.0 mm with 60 pixels is acquired. What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?
 - (b) One year later the same patient is scanned again, this time with an *axial* acquisition with voxel size in-plane $0.8 \times 0.8 \text{ mm}^2$ with a field-of-view of 192 mm in both directions, and voxel size through-plane of 2.0 mm with a field-of-view of 240 mm. What are the number of voxels in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?
 - (c) You want to compare a volumetric measurement of the same anatomical structure in the two images. Following the guideline given in the lecture, you resample the second image (b) onto the voxel grid of the first image (a). What are the resize factors along the head-to-toe, left-to-right and anterior-to-posterior directions, and what is the resulting image size? Why do you match the voxel size and not the image size?

2. Many noise suppression methods can be written as a weighted average $\hat{I}(x) = \sum_y w(x, y) J(y)$ of the noisy image J , with $w(x, y) \geq 0$ and $\sum_y w(x, y) = 1$. Which of the two weight definitions below is the one used by the *non-local means* filter? $N(x)$ denotes the image patch centred at x , $Z(x)$ is a normalisation constant and h the degree of filtering. Please circle the correct choice.

$$w(x, y) = \frac{1}{Z(x)} \exp \left(-\frac{\|J(x) - J(y)\|_2^2}{h^2} \right)$$

$$w(x, y) = \frac{1}{Z(x)} \exp \left(-\frac{\|J(N(x)) - J(N(y))\|_2^2}{h^2} \right)$$

3. Deformable intensity-based registration is usually written in one of the two equivalent forms below. The first is derived from a probabilistic model, the second is energy-based; T_θ is the transformation with parameters θ , I the target and J the moving image.

$$\max_{\theta} \log p(I \mid T_\theta \circ J) + \log p(T_\theta) \qquad \min_{\theta} d(I, T_\theta \circ J) + \lambda r(T_\theta^{-1})$$

- (a) Please indicate which term corresponds to which term in the two formulations.

- (b) Deformable registration is an ill-posed problem. Which of the two terms addresses this, and how?

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.

- (a) True False In the discrete setting, the *forward* transformation is preferred over the backward (inverse) transformation for resampling an image, because it guarantees that every pixel of the output grid is assigned a value.
 - (b) True False Nearest-neighbour interpolation is the appropriate choice for transforming a label (segmentation) map, because it does not introduce new values.
 - (c) True False Bilinear interpolation produces a function that is both continuous and differentiable.
 - (d) True False In 3D, Keys' cubic convolution (tricubic) interpolation uses only the 8 corner voxels of the cell that contains the query point.
5. Mixture models and Expectation-Maximization (EM) segmentation are essential building blocks for many segmentation algorithms. Consider a multi-modal image in which every pixel carries *two* intensity values (for instance a co-registered T1-weighted and T2-weighted pair). How many parameters does a Gaussian Mixture Model (GMM) with 3 components have? Count, for each component, the mean vector, the full covariance matrix and the mixture weight.
6. An Active Shape Model (ASM) is fitted to an image iteratively. Once the model has been initialised and candidate points have been found by sampling image features along the surface normals of the model points, each iteration updates the model in two parts. Please briefly describe what these two parts do, and state what keeps the fitted shape anatomically plausible. You do not need to write down the equations.
7. Thin-plate splines (TPS) and free-form deformations (FFD) with cubic B-splines are both interpolation-based models for non-linear transformations. Which of the following statements on their similarities and differences is **inaccurate**?
- ☐ (A) Both models place displacement vectors on a set of control points and interpolate in

between with a smooth basis function, so that the resulting transformation is smooth for any value of the parameters.

- ☐ (B) The influence of a single control point is global in TPS, whereas for FFD with cubic B-splines it is local, because the B-spline basis functions have finite support.
 - ☐ (C) In FFD, increasing the number of control points, i.e. using a finer control grid, results in smoother transformations.
 - ☐ (D) For a given set of N control points in 3D, both models are parameterised by the $3N$ coefficients of the displacement vectors at the control points.
8. Multi-atlas and patch-based methods were the state of the art in segmentation for a long time. Which of the following statements about them is **incorrect**?
- ☐ (A) In multi-atlas segmentation, every atlas is registered to the unseen image, each atlas produces its own propagated segmentation, and these are combined in a decision fusion step such as majority voting or STAPLE.
 - ☐ (B) STAPLE estimates the most probable underlying true segmentation and, simultaneously, the sensitivity and specificity of each of the input segmentations; it is solved iteratively with an Expectation-Maximization algorithm.
 - ☐ (C) Selecting only the atlases that are similar to the unseen image before fusing performs worse than naively fusing all available atlases, because discarding atlases throws away information.
 - ☐ (D) Patch-based label fusion requires only a coarse, rigid or affine registration, because correspondence is re-established locally by comparing patches inside a search window.
9. Convolutional neural networks and Vision Transformers (ViT) build very different assumptions into the model. Which of the following statements is **incorrect**?
- ☐ (A) In CNNs, locality and the two-dimensional neighbourhood structure of the image act as strong inductive biases for vision tasks.
 - ☐ (B) In a ViT, self-attention is global and only the MLP layers act locally, so that splitting the image into patches is essentially the only two-dimensional inductive bias built into the model.
 - ☐ (C) Because a ViT builds in fewer assumptions about images, it typically requires *less* training data than a comparable CNN to reach the same accuracy.
 - ☐ (D) Without positional encodings, a transformer encoder is equivariant to permutations of its input tokens, which is why positional information has to be added to the token embeddings.
10. You are training a U-Net to segment a small lesion that occupies far less than 1% of the image. Which of the following statements is correct?
- ☐ (A) With pixel-wise cross-entropy, the loss contributed by the lesion is typically much smaller than the loss contributed by the rest of the image, so the lesion can be dominated during training.
 - ☐ (B) The Sørensen-Dice coefficient computed on the binary label sets is differentiable and can therefore be used directly as a training loss.
 - ☐ (C) The Dice loss makes the contribution of a class proportional to its size, which is why it helps for small structures.

- (D) The skip connections of the U-Net remove any need to reconsider the loss function, because they restore the resolution lost in the encoder.
- (E) Multi-class cross-entropy cannot be used when more than two classes are present.
11. Consider the encoder of a segmentation network that takes grey-scale input images of size 128×128 with one input channel and has the following architecture:
- **Conv1**: 32 kernels with kernel size $K = 3 \times 3$, stride $S = 1 \times 1$ in each direction, and “same” padding $P = 1$.
 - **Pool1**: Max pooling with $K = 2 \times 2$ and $S = 2$.
 - **Conv2**: 64 kernels with $K = 3 \times 3$, $S = 1$ and “same” padding $P = 1$.
 - **Pool2**: Max pooling with $K = 2 \times 2$ and $S = 2$.
 - **Conv3**: 128 kernels with $K = 3 \times 3$, $S = 1$ and “same” padding $P = 1$.
 - **Pool3**: Max pooling with $K = 2 \times 2$ and $S = 2$.

Circle True or False:

- (a) True False The global receptive field of a neuron in Pool3 is 22×22 .
- (b) True False The global stride of a neuron in Pool3 is 3×3 , one for each pooling layer.
- (c) True False The total number of convolutional kernel weights (excluding the biases) in Conv1, Pool1, Conv2, Pool2, Conv3 and Pool3 is $3 \cdot 3 \cdot 1 \cdot 32 + 3 \cdot 3 \cdot 32 \cdot 64 + 3 \cdot 3 \cdot 64 \cdot 128 = 92\,448$.
- (d) True False The output of Pool3 has a spatial size of 16×16 , and each of its neurons therefore sees the entire 128×128 input image.
12. Which of the following statements on uncertainty estimation, domain shift and learning from unlabelled examples are correct (multiple answers are possible)?
- (A) Standard, deterministic training and inference amount to approximating $p(\theta \mid D, M)$ by a Dirac delta at θ^* and $p(y \mid x, \theta, M)$ by a Dirac delta at $f(x; \theta)$, so that the resulting predictive distribution is a point mass carrying no uncertainty at all.
- (B) Test-time adaptation still requires labelled samples from the target domain, but fewer of them than fine-tuning does.
- (C) Self-training with pseudo-labels works well only if the predictions of the initial model are reasonably good; with poor initial estimates it can diverge quickly.
- (D) In the mean-teacher model the teacher is trained by minimising the supervised loss on the labelled examples, while the student is the exponential moving average of the teacher.
- (E) Domain generalisation by extreme data augmentation trains only in the source domain, but the improvement it brings may be limited to the kinds of variation that the augmentations actually cover.